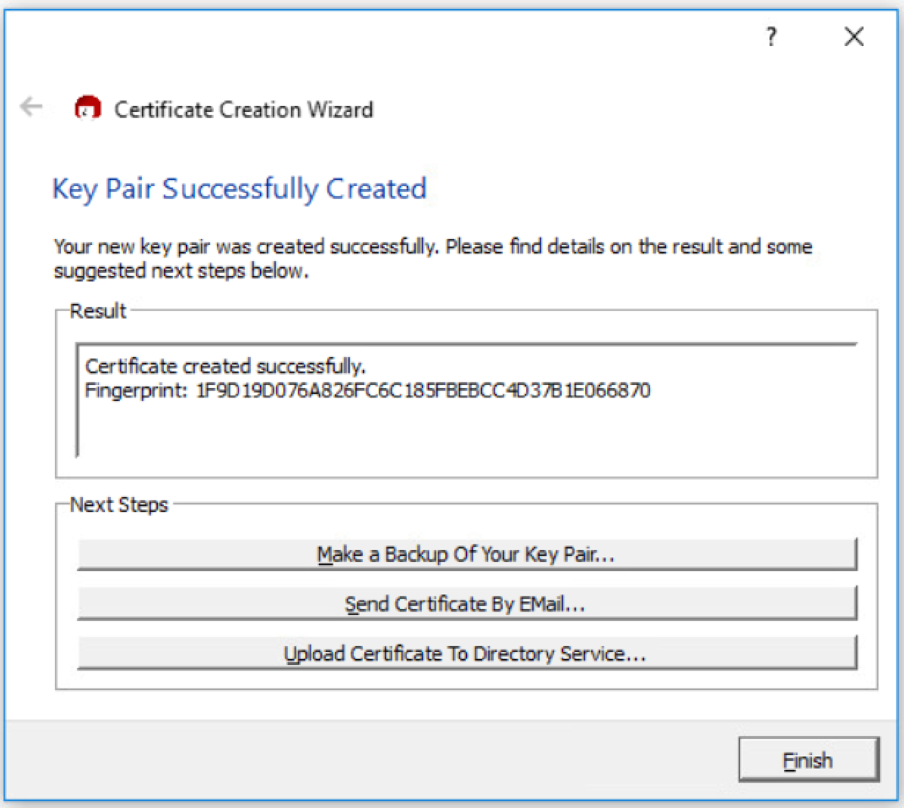
Lab 8.1: Using Public Key Encryption to Secure Messages

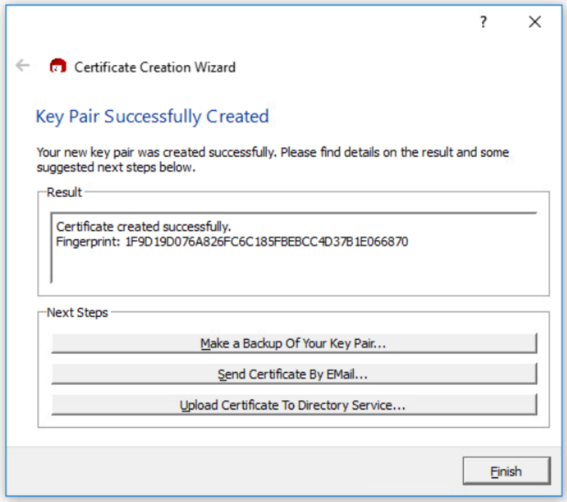
1. Research and describe what **Kleopatra** is

Replace with written response

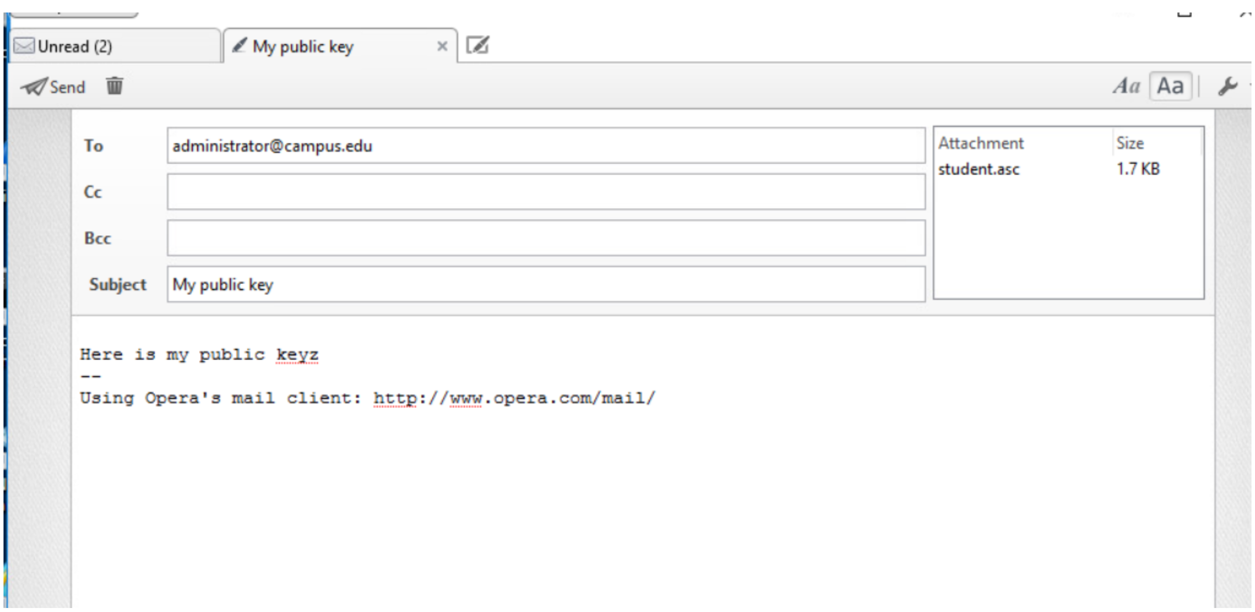
2. Question 5



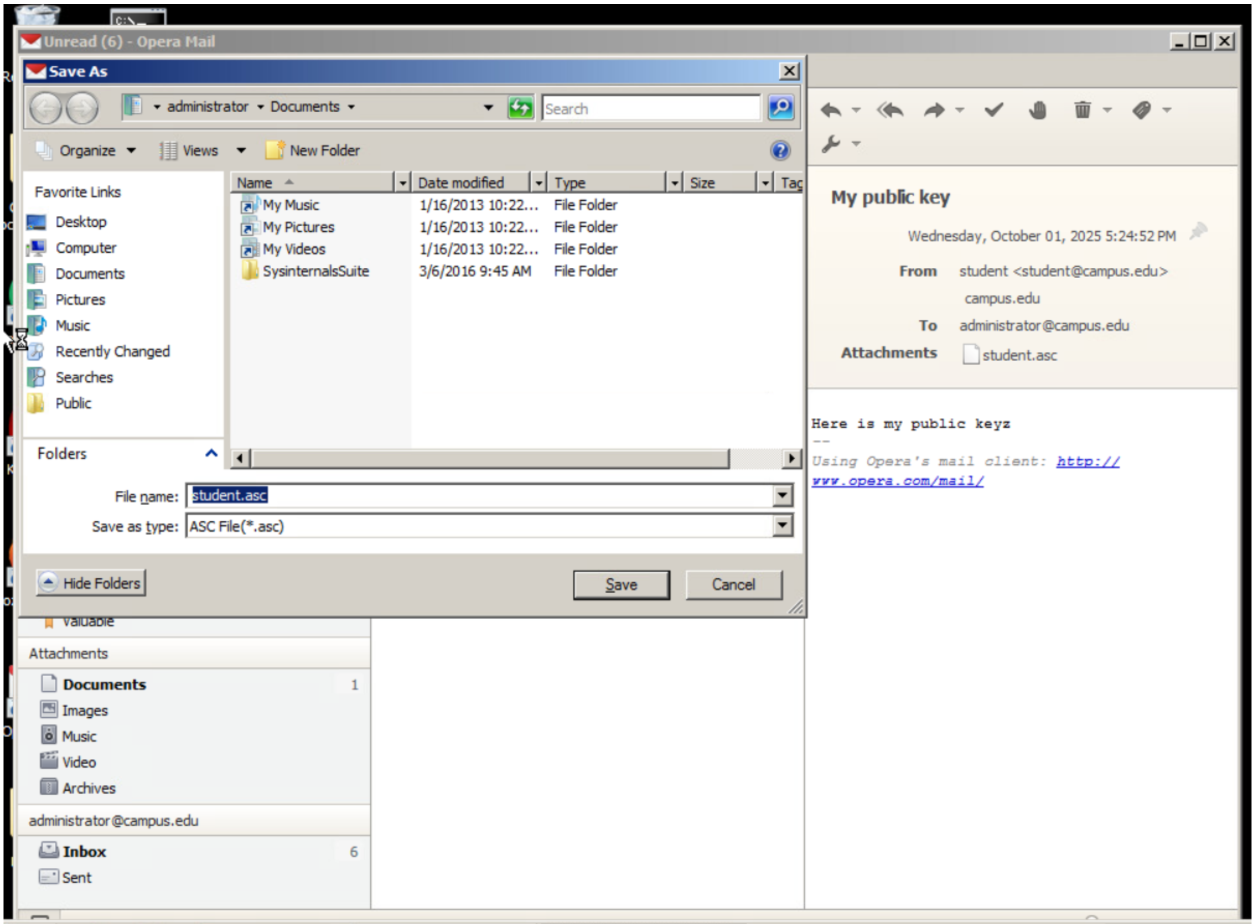
3. Question 10



4. Question 20

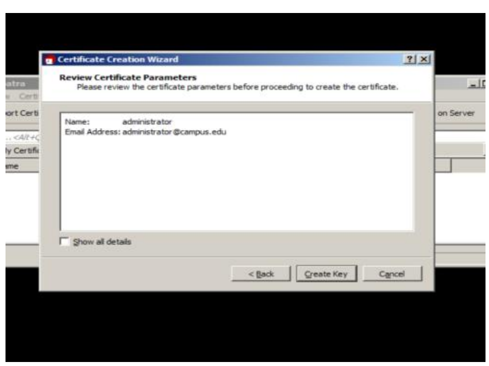


5. Question 28

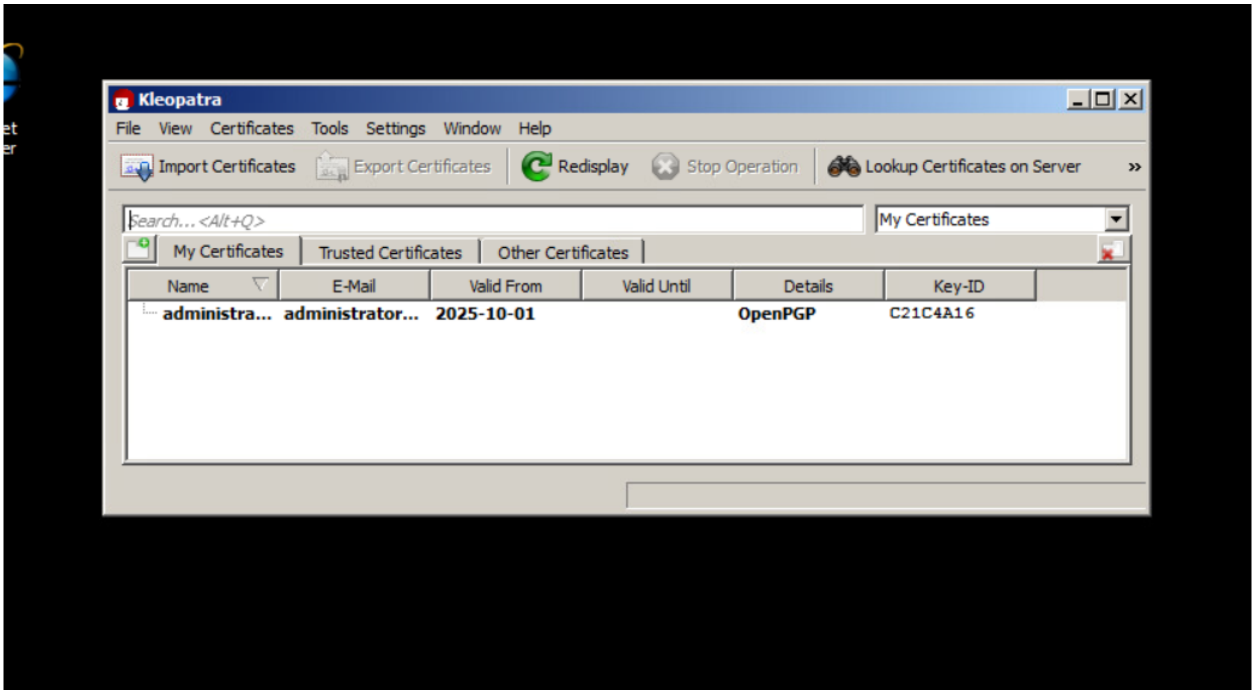


Lab 8.1: Part 2 – Creating the Certificate for Administrator

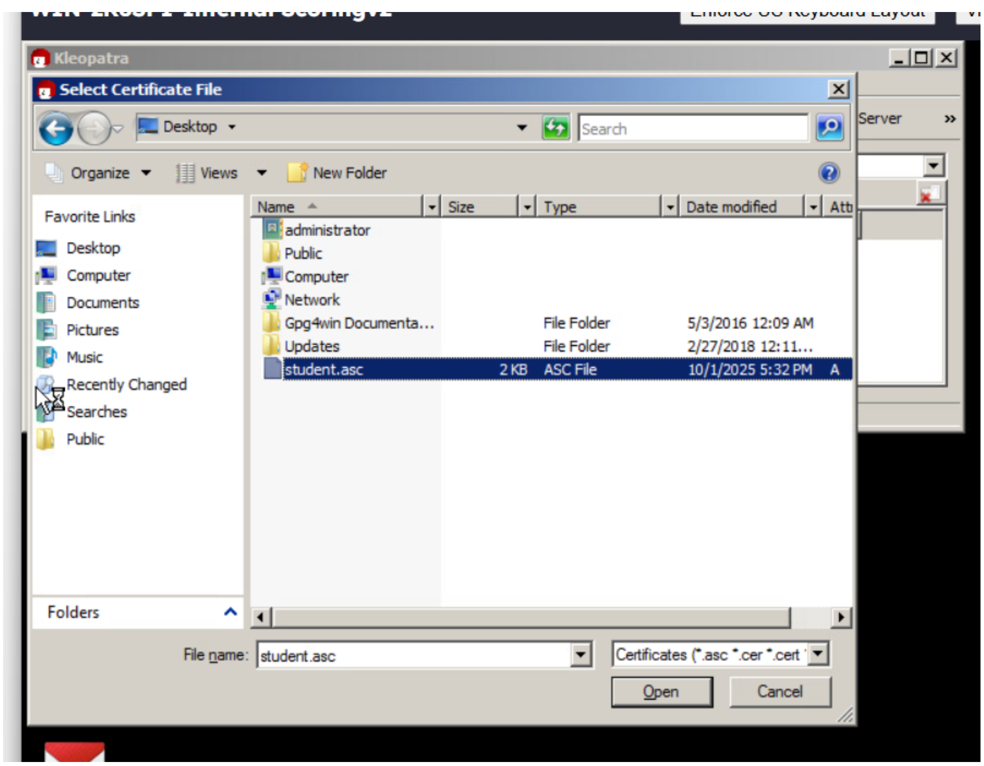
1. Question 5



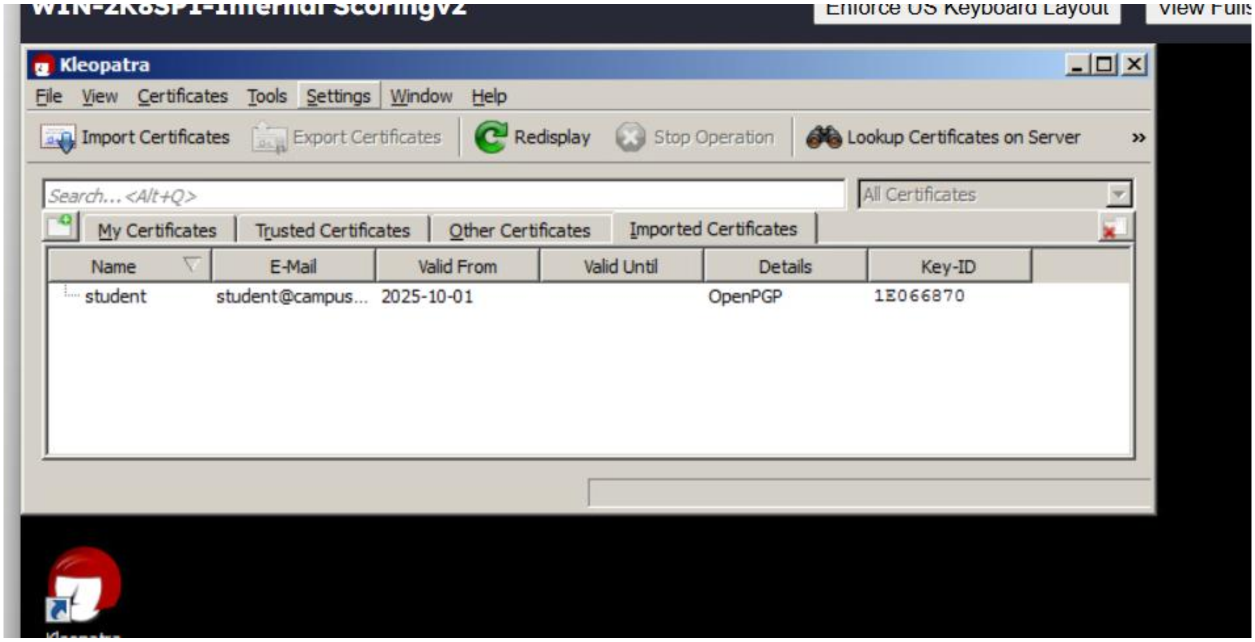
2. Question 9



3. Question 12

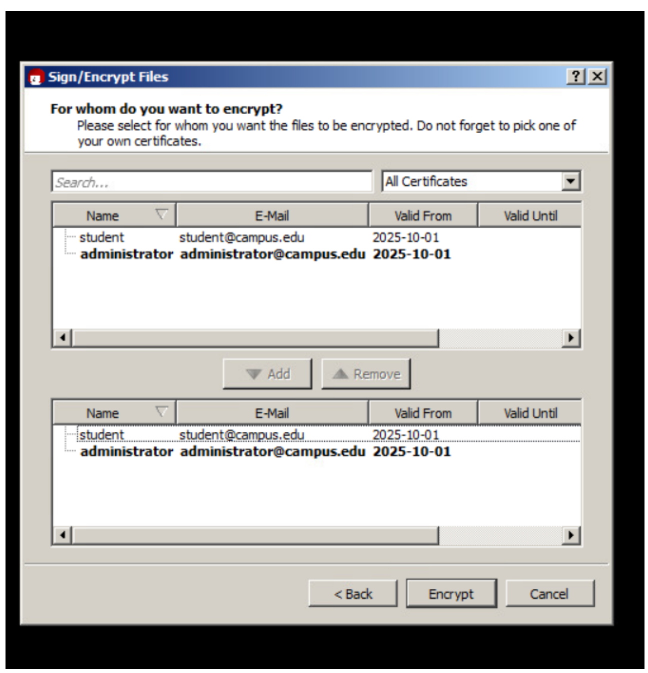


4. Question 13

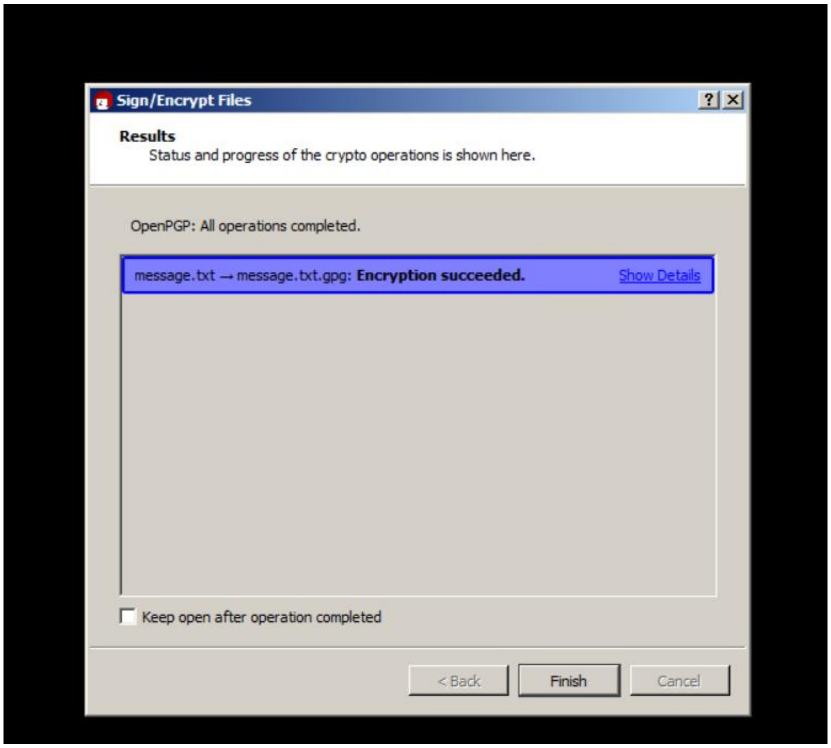


Lab 8.1: Part 3 – Encrypting and Decrypting the File

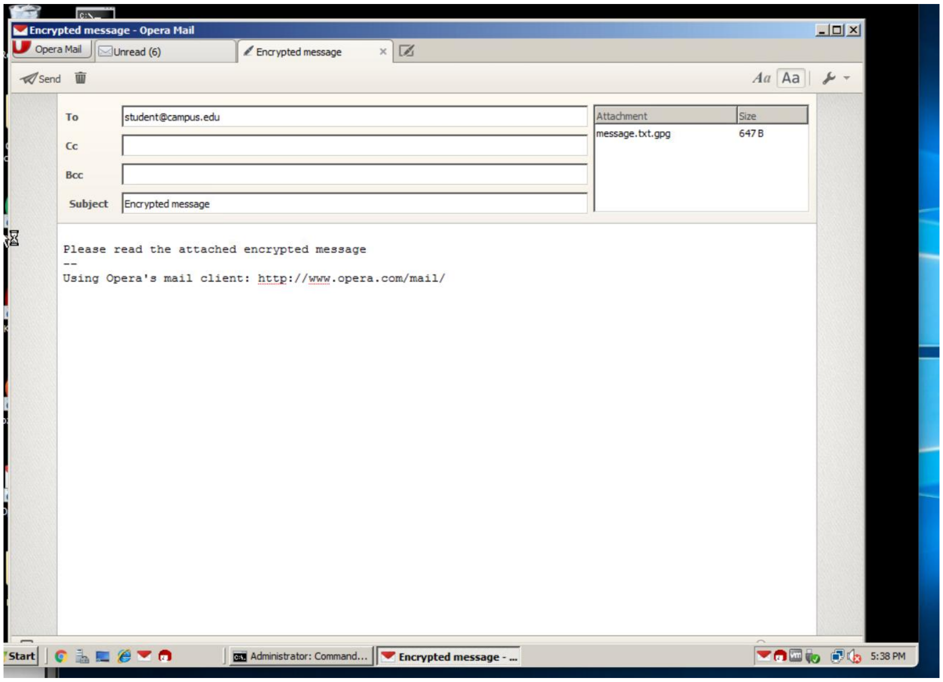
1. Question 10



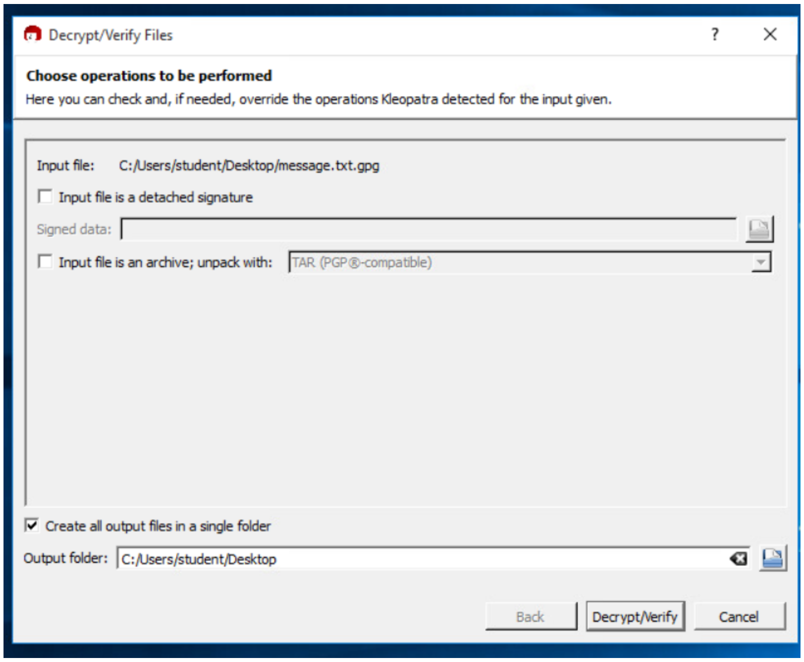
2. Question 11



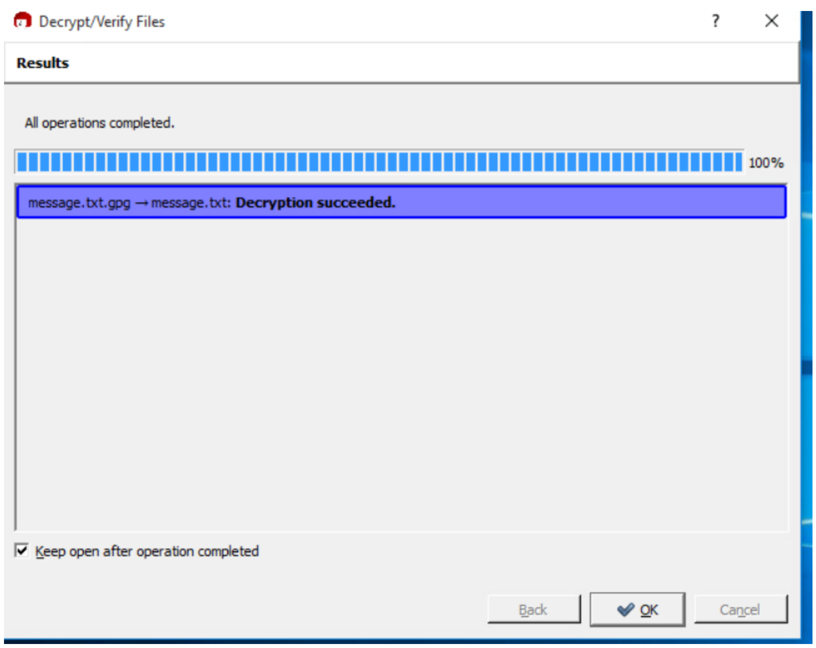
3. Question 16



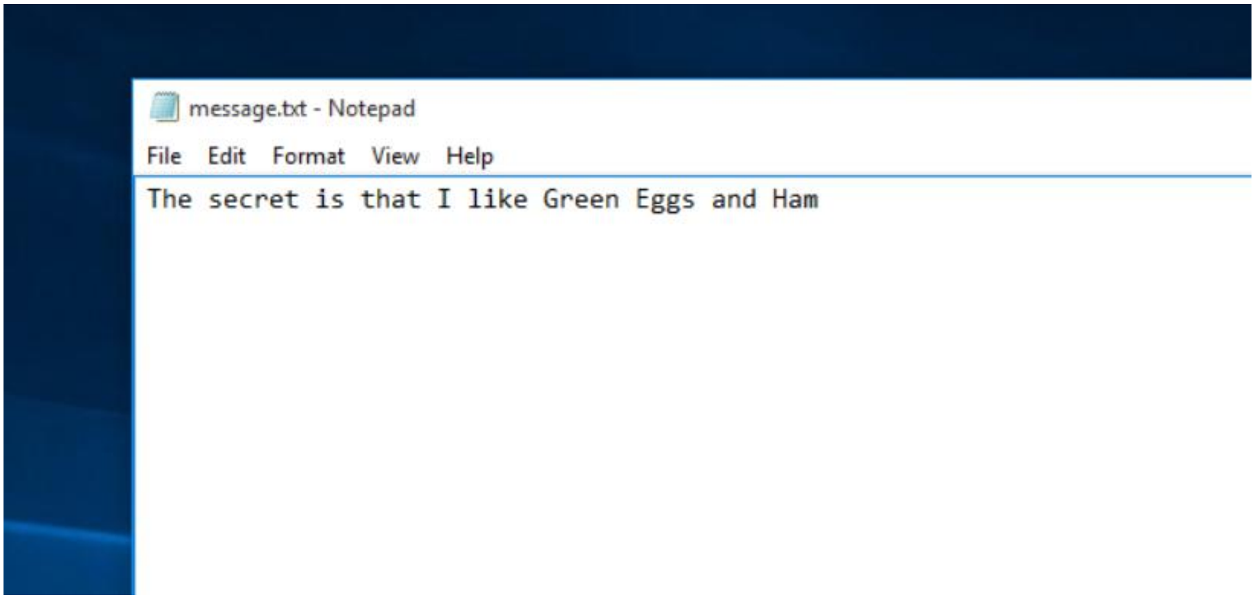
4. Question 24



5. Question 26



6. Question 28



Lab 8.1: Part 4 – Lessons Learned

This lab provided hands-on experience with public key encryption, specifically using tools like Kleopatra to create and manage digital certificates. I learned how asymmetric encryption works in practice, including the generation of key pairs and the process of encrypting and decrypting files securely. Understanding the role of a certificate authority and how trust is established between users was particularly insightful. The lab helped reinforce theoretical concepts from class by allowing me to see how encryption protects data confidentiality and integrity in real-world scenarios. One of the surprising aspects of the lab was how user-friendly Kleopatra is despite dealing with complex cryptographic operations.